Q31. Has NRCP released any new variety of pomegranates?

Two hybrid varieties 'Solapur Lal' for table and processing purpose and 'Solapur Anardana' for processing have been released in 2017.

Q32. How is Super Bhagwa different from Bhagwa?

Super Bhagwa matures 15-20 days early to Bhagwa and has better size and improved colour than Bhagwa, however, Super Bhagwa at NRCP and some farmers fields was not found to be significantly different from Bhagwa.

Q33. What are your views on the performance of Variety "Wonderful" in India?

"Wonderful" is one of the oldest variety of Pomegranate from USA. It has many variants/landraces spread across the temperate as well tropical world (Wonderful, Pome Wonder, Early Wonderful, Israeli Wonderful, P.G.101-2, etc). Temperate types 'Wonderful' do not flower in hot arid and semi arid regions of India. Hence not suitable for cultivation in these areas. Nevertheless, the variety 'Wonderful' has very large fruit size weighing around 500 g, TSS 17-18°Brix attractive red peel and arils but high acidity (1-1.5% in comparison to only 0.4 % in Bhagwa) and has hard seeds compared to soft seeded 'Bhagwa'. It is suitable for processing.

Q34. Is Wonderful better than Bhagwa? Is it resistant to bacterial blight?

Wonderful is not suitable for Indian markets as Indians like sweet taste and soft seed like that in Bhagwa. The Wonderful plants at NRCP now more than 2 years old have not flowered so far but foliage is highly susceptible to bacterial blight.

Q35. What is the recommended spacing for pomegranate plantation?

Spacing of 4.5 m (15 ft.) X 3 m (10 ft.) which accommodates about 296 plants per acre is recommended. Trials for HDP are in progress at NRCP.

Q36. What intercrops can be taken and which should be avoided in pomegranate?

Sunhemp and other green manure crops help improve natural beneficial micro flora and should be preferred as inter-crop and later buried in soil. Maize, wheat, sorghum, bajra, mustard also reduce nematode population. Planting *Tagetes erecta* (African marigold) varieties 'Pusa Basanti Gainda' and 'Pusa Narangi Gainda' for continuously 6-7 months is beneficial for reducing nematode population in infested orchards. For effective results these should be grown for at least 6-7 months continuously.

Crops like onion, tomato, chili, potato, capsicum, carrot, gram, legumes (tur, urd, lentil, rajmash, beans, soybean etc., cucurbits (cucumber, melons), Gerbera, Gladiolus etc. aggravate nematode infestation and hence should be avoided as intercrop. Cucurbit crops increase insect problems also, which in turn transmit and spread several pathogens, hence should be avoided. Several fruit crops are hosts of fruit borer (*Deudorix isocrates)* and other insect pests, hence avoided. Similarly thrips are common on horticultural crops vegetables, ornamentals as well as fruits, hence, avoid taking them as inter-crops. Vegetables and ornamentals serve hosts to several insect pests and should in general avoided as inter-crops.

Q37. What is the perfect pruning procedure to get good flowering and fruit set? How many times in a year pruning should be practiced?

Prune twice a year to maintain proper canopy and fruit set (i) Main pruning just after harvest (ii) Light (upper 10-15 cm) pruning at flower regulation. Never prune during rainy season. Remove all shoots upto 30-60 cm above ground depending on plant height. Avoid too heavy pruning, confine pruning during dormant stage to previous year growth. Allow set of new shoots to develop every year on all sides of the tree.

Q38. When to do harvesting of pomegranate fruits?

As pomegranate fruits are non-climacteric in nature, the fruits should be harvested after attaining maturity on the plants. The maturity period from fruit set is about 6 months for Bhagwa and 5 months for Ganesh. Depending on the climatic conditions and management practices the period may vary by few weeks.

Q39. What operations have to be followed during rest period?

The recommended fertilizer doses (as in Q18) have to be applied and light irrigation has to be given to maintain proper rest conditions, Heavy pruning has to be done soon after harvest. Based on the requirement insect pests and diseases management schedule has to be followed. Otherwise atleast Bordeaux mixture 1% or other copper sprays should be taken at 10-15 days interval.

Q40. When should we start training of new planation?

Ideally plants are trained between 3-6 months or when they are around 3ft. height with good growth.

Q41. What are pruning procedures for plants of different age?

In new plantation, during pre-bearing period (upto 2 year), the juvenile plant may be pruned by removing the criss-cross branches, water shoots etc. Pruning is also done to regulate flowering and fruiting of pomegranate plants of bearing orchard (Above 2 years age). In general, a light pruning is given by removing 5-6 inches of shoot tips at flower regulation. After harvesting, dead and dried shoots, criss-cross branches, water suckers are removed & basal fertilizers dose is applied for rest period.

IV. SOIL AND NUTRITION

Q42. We have black cotton soil, is it advisable to grow pomegranate?

No, it is not advisable to grow pomegranate in black cotton soil. If you don't have any alternative, you need to apply high amounts of organic manure and need to have provision for easy drainage. For induction of flowering, you need to withhold irrigation for longer period (say 3 to 4 months) of time.

Q43. Can I do amendments to make black soil type favorable for pomegranate?

Though not advisable to grow pomegranate in black soil, yet if you want to grow in black soil, add sufficiently high quantity (40Kg/plant/year) of organic manure and grow green manure crops every year, followed by mixing with soil in presence of sufficient moisture for decomposition. This will improve the porosity of soil there by improving the drainage condition of soil. Put plants to stress for at least 3-4 months in order to induce good flowering.

Q44. I am giving all nutrients but plant growth is not good. What should I do?

You should get your soil tested for pH and EC. If the pH is above 8.5, you need to check exchangeable sodium percentage (ESP), if it is more than 13% it indicates sodicity of soil, then you need to apply Gypsum according to gypsum requirement (estimated in laboratory) through soil analysis followed by leaching with good quality water. if the soil is not sodic, then you should apply sufficiently high quantity of well decomposed organic manure /humic acid etc.

If roots have nematode infestation then also plant growth may be affected, hence, suitable nematode control treatments should also be followed.

Q45. Tell me easiest way to understand the type of soil I have and methods to rectify the problematic soils.

Moistened the soil slightly and roll it into a ball on your palms, if-

(i) No ball formed and soil crumbles through fingers easily, it is a **sandy soil**. Add lot of organic manure.

(ii) Soil feels sticky, rolls up easily, and forms into a ball, then it is a **clay soil**. Add 50% sand and organics and mix.

(iii) Smooth, partly gritty, partly sticky ball that crumbles easily is **loam soil**. This is ideal soil type for planting add recommended organic and nutrients

Among other soil types are:

Saline soil: A white layer/coating on the surface of the soil below drippers means you have saline soils. Leaf tip burn, especially on young leaves also is seen in saline soils. Leaching of salts with irrigation water and planting should be done on the middle position of slant surface of ridge and not on the top of ridge.

Sodic soils: Soil containing excess exchangeable sodium (Na) are sodic soils, the soil becomes water logged and later when it dries you can see upper layer of soil peeling off in thin layers. Application of gypsum at 50% of gypsum requirement (assessed through laboratory analysis) followed by leaching of salt with irrigation water is recommended.

Q46. What is the proper method for fertilizer application?

Apply fertilizers in ring during rainy season and below drippers in other seasons.

How can we increase Calcium (Ca) content in fruit?

In calcareous soil

1. Apply sufficient quantity well decomposed farmyard manure followed by irrigation.

2. Apply elemental S @ 20-30 g per plant in root zone and thoroughly mix with the soil.

In non-calcareous soil

1. Apply gypsum @ 500 700 g per plant depending on the pH of soil.

Q48. Can we apply Calcium and Boron together to prevent fruit cracking?

For best results boron should be applied through foliar spray while soil application of gypsum is most effective in enhancing calcium content in fruit.

Q49. I have applied all nutrients still the fruit size is not increasing. The skin appears hard. What should I do to?

Sometimes, fruit skin get hardened due to excessive use of copper based pesticides especially in acidic PH, or due to intense heat, low humidity and irrigation water. Always check pH of ready to use spray solution which should be between 6.5-7.0 pH. All copper sprays should be done at 7 pH You can go for three foliar application of KH_2PO_4 @ 10g/l and $MnSO_4$ @ 6 g/l at 15 days interval starting from 60 days after flowering for increasing the fruit size.

Q50. Is fertigation useful under saline soil conditions?

Fertigation should be avoided under saline soil condition in order to prevent further salinization. Use of water soluble fertilizer enhances soil salinity.

Q51. How can we reduce soil pH in the rhizosphere for enhancing availability of cationic micronutrients?

Use of gypsum in non-calcareous alkaline soil and elemental S in calcareous alkaline soil according to gypsum requirement will reduce localized soil pH and enhance availability of micronutrients. (See answer to Q.47)

Q52. How to reclaim saline soil for better plant growth?

Saline soils can be reclaimed by leaching salt from the root-zone with poor quality irrigation water diluted with good quality fresh water in 2:1 ratio and using 60-70 kg well decomposed farmyard manure per plant. Growing of plant on the middle of the slant of inverted V-shaped ridge avoids salt accumulation in the root-zone.

Q53. Can micronutrients be applied through soil?

If soils are found to be deficient in particular micronutrients, those micronutrients can be mixed with well decomposed farmyard manure and incubated for 14 days in shade with slight humidity, then it can be applied to the root-zone for reclaiming micronutrient deficiency.

Chelated Fe (Fe-EDDHA) can be used for soil application in high pH soil while other micronutrients like Zn, Mn, Cu and B are more effective in foliar application of their inorganic water soluble salts (i.e. $ZnSO_4$, $MnSO_4$, $CuSO_4$, and H_3BO_3 respectively)

Q54. Is EDTA chelate of Zn or other fertilizers more effective than inorganic salt of Zn ($ZnSO_4$)?

No, foliar application of inorganic salt of Zn is more effective in enhancing Zn content of fruit than EDTA chelated Zn. Same is true for other fertilizers. The size of EDTA molecule is bigger than stomatal aperture; hence its absorption is prevented through plant surfaces.

Q55. Is slurry application beneficial?

Slurry application is highly beneficial if it is made from well decomposed cow-dung manure, as it facilitates better penetration of nutrient and other growth promoters to the root-zone.

Slurry of fresh cow-dung should not be used as it immobilizes nutrients from the rhizosphere As a result,

Q56. Are inorganic fertilizers needed even if I am using lot of organic fertilizers?

Yes, it is needed in order to have higher productivity. With only organic fertilizers, it is very difficult to maintain high productivity as they contain low amount of nutrients per unit weight as compared to inorganic fertilizers, integrating inorganic fertilizers with organics is the best option for enhancing productivity and quality.

Q57. Is excessive application of nitrogen harmful?

Yes, excessive application of N is harmful to plant. High nitrogen promotes excessive plant vegetative growth, so the plant produces less number of flowers and fruits. Moreover, plants become more susceptible to diseases and insect pests especially bacterial blight.

Q58. Which nitrogen fertilizers should be used under different soil types?

Neem coated urea can be used in calcareous soil. However, ammonium sulphate is the best source of N for pomegranate although it is costly. In non-calcareous soil Calcium ammonium nitrate (CAN) can also be used.

Q59. How nitrogen (N) can be managed in pomegranate?

Nitrogen should always be applied in split doses. One-third N of recommended dose as per age should be applied just after harvest of fruit in previous cropping and remaining 2/3rd can be applied in two equal split one at 60 days after full bloom and another at 120 days after full bloom.

Q60. Can we use zinc sulphate ($ZnSO_4$), manganese sulphate ($MnSO_4$), boric acid and bactericide together for foliar application?

Yes, they can be used together for foliar application without any interference of each other.

Q61. Is potash (K) application required in soil rich in K?

Yes, you need to apply potash after every harvest as per soil and leaf analysis test reports. Pomegranate removes maximum of K with the harvest, so it should be replenished in order to maintain sustainability of production. Moreover, applied K does not get lost from the system as with N. So it will be available to the plant in due course of time.

Q62. Which source of K is good for improving storage life and quality of fruit?

Sulphate of potash (K_2SO_4) is the good source of K in pomegranate for improving storage life and quality of fruit.

Q63. What is the role of salicylic acid in pomegranate? How and how much has it to be used?

Salicylic acid is an important plant defense molecule involved in multiple physiological processes. It imparts resistance to plants against many diseases. In pomegranate only 4 applications of 300 ppm (0.3g/l) at 30-40 days interval starting from pre flowering are advised. If higher doses are used or more applications are taken, then it will disturb and affect other physiological pathways involved in plant growth and development.